SEFU HMIS — Comprehensive Business Continuity Plan (BCP)

Kenya Digital Health Agency (DHA) Certification Compliance Document

Category: Information Security Documentation — Section C: Business Continuity

System Classification: Health Information System (HIS) & Electronic Medical Record (EMR)

Applicable Regulations: Kenya Digital Health Act 2023, Kenya Data Protection Act 2019, ISO 22301 (Business Continuity Management), Kenya National e-Health Standards

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■ 1. Executive Summary & Document Control

Attribute	Specification
Document Title	Business Continuity Plan (BCP) & Clinical Downtime Operations Specification
System Name	SEFU Hospital Management Information System (SEFU HMIS)
System Version	Release 5.0 (Build 2026-Kenya-DHA)
Document Version	1.0 (Production / Certification Ready)
Document Owner	Chief Technology Officer & Hospital Medical Superintendent
Effective Date	November 2026
Review Frequency	Annually or following significant hospital operational restructuring
Classification	Confidential — Healthcare Operational Resilience Documentation

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■ 2. Purpose, Scope & Objectives

2.1 Purpose

The purpose of this Business Continuity Plan (BCP) is to ensure that healthcare delivery, patient safety, and clinical workflows continue without interruption during planned or unplanned outages of the digital HMIS infrastructure. While the Disaster Recovery Plan (DRP) focuses on technical server and database restoration, the BCP focuses on **uninterrupted clinical care at the patient bedside**.

2.2 System Scope

This plan governs all hospital units operating SEFU HMIS:

- **Emergency Department & Trauma Unit**
- **Outpatient Clinics (OPD) & Specialty Services** (Pediatrics, Physiotherapy, Gynaecology)
- **Inpatient Care Units (IPD)** (ICU, Maternity, Pediatric, Male/Female Surgical & Medical Wards)
- **Diagnostic Facilities** (Pathology Laboratory, Radiology Imaging)
- **Hospital Pharmacy & Medical Consumables Store**
- **Cashier, Billing & Health Insurance Desk** (Social Health Authority / SHA & Private TPA)

■ 3. Business Impact Analysis (BIA) & Clinical Criticality

Healthcare workflows cannot tolerate prolonged system unavailability without threatening patient clinical outcomes. The following table identifies core hospital services and their Maximum Tolerable Outage (MTO):

Critical Clinical Process	MTO (Maximum Tolerable Outage)	Clinical Risk if Interrupted	Fallback Continuity Procedure
Emergency Triage & Resuscitation	**0 Minutes (Immediate)**	Life-threatening delay in trauma / triage treatment.	Immediate switch to red-tagged manual paper emergency admission forms.
Inpatient Medication Administration	**15 Minutes**	Delayed or duplicate medication dispensing, missed doses.	Pre-printed paper Medication Administration Records (MAR) kept on ward trolleys.
Diagnostic Lab & Radiology Reporting	**30 Minutes**	Inability to access stat blood gas, cardiac markers, or imaging.	Verbal telephone / runner authorization followed by manual paper lab requisition.
Operating Theatre Scheduling & Surgery	**30 Minutes**	Surgical delay, anaesthetic record lapse.	Physical surgical whiteboards and paper operative logbooks.
Outpatient Consultation & Intake	**1 Hour**	Patient congestion, delayed consultations.	Paper OPD encounter slips; sequential manual queue tokens.
Discharge Billing & Cashier Desk	**2 Hours**	Delayed inpatient discharge, billing backlog.	Manual carbonized receipt books; deferred final insurance settlement.

■ 4. Clinical Downtime Procedures (Paper-Based Contingency)

When SEFU HMIS is unavailable (due to network disruption, server maintenance, or hardware fault), hospital staff immediately activate the **Clinical Downtime Packets**.

4.1 Physical Downtime Stationery Boxes

Every nursing station, clinic, pharmacy, and cashier desk maintains a dedicated **Downtime Stationery Box** containing pre-printed, standardized paper forms:

1. **Form DT-01:** Emergency & OPD Patient Registration Intake Voucher.
2. **Form DT-02:** Clinical Consultation & Progress Notes Sheet.
3. **Form DT-03:** Medication Administration Record (MAR) & Prescription Voucher.
4. **Form DT-04:** Diagnostic Laboratory & Radiology Requisition Slip.
5. **Form DT-05:** Inpatient Nursing Kardex & Vitals Flowsheet.
6. **Form DT-06:** Manual Official Hospital Cashier Receipt (Triplicate).

4.2 Offline Patient Identification Protocol

To avoid identity confusion during digital downtime:

- Temporary downtime patient IDs follow the format: `DT-[YYYYMMDD]-[WARD]-[SEQ]` (e.g., `DT-20261114-EMR-001`).
- Physical wristbands bearing the temporary downtime number and full patient name are placed on all admitted patients.

■ 5. Physical Infrastructure & Redundancy Safeguards

5.1 Power Resilience (Tier 1 Hospital Power Grid)

- **Uninterruptible Power Supplies (UPS):** All central server racks, nurse station terminals, and network switches are connected to enterprise online double-conversion UPS units providing a minimum of **45 minutes** battery runtime.

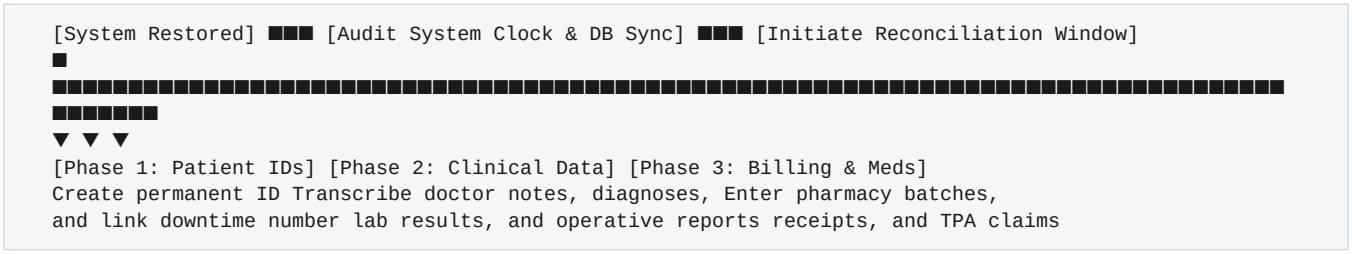
- **Automated Diesel Generator:** The hospital facility is backed by dual automated backup generators with an automatic transfer switch (ATS) activating within **12 seconds** of utility power failure.
- **Fuel Reserves:** On-site diesel storage guaranteed for a minimum of **7 days** continuous operation.

5.2 Network & Internet Redundancy

- **Dual ISP Routing:** Primary high-speed Dedicated Optical Fiber supplemented by automated failover to an enterprise 5G / LTE cellular gateway.
- **Local LAN Survivability:** Internal hospital switches and local Wi-Fi routing remain fully operational even if external internet connection to the cloud or ISP is severed, allowing local server communication.

■ 6. Digital Post-Downtime Data Reconciliation (The "Catch-Up" Runbook)

Once SEFU HMIS digital services are fully restored, paper records accumulated during the outage must be systematically transcribed into the database to restore unified digital records:


[System Restored] ■■■ [Audit System Clock & DB Sync] ■■■ [Initiate Reconciliation Window]
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■ 7. Business Continuity Command Structure

Role	Person Responsible	Core Continuity Duties
BCP Director	Chief Executive Officer (CEO) / COO	Authorizes downtime declaration, interfaces with Ministry of Health & Media.
Clinical Operations Commander	Medical Superintendent	Coordinates physicians, ensures patient clinical safety, monitors bed capacity.
Nursing Continuity Coordinator	Chief Nursing Officer (CNO)	Mobilizes ward downtime stationery, ensures manual MAR enforcement.
IT Continuity Lead	CTO / Lead Systems Administrator	Executes technical recovery, coordinates ISP/power engineers, leads reconciliation.
Communications & Legal Officer	Hospital PR / Data Protection Officer	Coordinates staff broadcasts, ensures patient data protection compliance.

■ 8. Training, Drills & Maintenance Schedule

- **Staff Onboarding:** Every new clinical and administrative employee receives mandatory training on physical downtime packets.
- **Semi-Annual Tabletop Drills:** Every 6 months, hospital units conduct a simulated 2-hour digital blackout drill to test paper charting, triage speed, and reconciliation accuracy.
- **Annual BCP Audit:** Review of stationery supplies, UPS battery degradation tests, and revision of clinical contact directories.

Approved by:

Hospital Medical Superintendent & Chief Technology Officer

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